



FRS

Functional Requirements Specification

PROJECT DETAILS

Project: TravelMate

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1. Introduction

This Functional Requirements Specification (FRS) defines the functional and non-functional requirements for the TravelMate Minimum Viable Product (MVP). It provides a shared understanding among business stakeholders and the project team regarding the system functionality, project scope, business rules, quality attributes, and traceability between the Business Requirements Document (BRD) and the Functional Requirements Specification (FRS).

2. Purpose

- Define the functional behavior of the TravelMate MVP.
- Describe how the system shall support business requirements.
- Provide functional specifications for development and testing.
- Serve as a reference for developers, testers, and business stakeholders throughout the project lifecycle.

3. Scope

Account Management:

- User Registration
- User Authentication
- Profile Management
- Password Management

Trip Management:

- Trip Creation
- Trip Maintenance
- Trip Archiving
- Trip Restoration
- Trip Deletion

Collaboration Management:

- Collaboration Enablement
- Member Invitation
- Member Management
- Role Management

4. Functional Requirements

Account management:

ID	Functional Requirement	Priority
FR-001	The system shall allow travelers to register an account.	Must
FR-002	The system shall validate required registration information.	Must
FR-003	The system shall prevent duplicate email addresses.	Must
FR-004	The system shall validate user credentials before granting access.	Must
FR-005	The system shall allow travelers to log out securely.	Must
FR-006	The system shall allow travelers to edit profile information.	Must
FR-007	The system shall allow travelers to change passwords.	Must

Trip management:

ID	Functional Requirement	Priority
FR-008	The system shall allow travelers to create trips.	Must
FR-009	The system shall allow travelers to edit trips.	Must
FR-010	The system shall allow travelers to archive trips.	Must
FR-011	The system shall allow travelers to restore archived trips.	Must
FR-012	The system shall allow travelers to delete trips.	Must
FR-013	The system shall display the traveler's trip list.	Must

Collaboration:

ID	Functional Requirement	Priority
FR-014	The system shall allow Trip Owners to enable collaboration.	Must
FR-015	The system shall allow Trip Owners to invite existing users via email.	Must
FR-016	The system shall allow invited users to accept invitations.	Must
FR-017	The system shall allow invited users to decline invitations.	Must
FR-018	The system shall allow Trip Owners to remove members.	Must
FR-019	The system shall allow Trip Owners to change member roles.	Must
FR-020	The system shall display trip members and their roles.	Must
FR-021	The system shall record collaboration activities.	Must
FR-022	The system shall send email notifications for collaboration events.	Must

5. Non-functional Requirements

ID	Requirement
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NFR-001	System response time shall not exceed 3 seconds.
NFR-002	System availability shall be at least 99%.
NFR-003	Passwords shall be encrypted before storage.
NFR-004	Only authenticated users may access protected resources.
NFR-005	Email notifications shall be delivered asynchronously.
NFR-006	Trip data shall be stored securely.
NFR-007	System communications shall use HTTPS.
NFR-008	Only authorized users shall access trip information.
NFR-009	The system shall support concurrent access by multiple users.

6. Business Rules

ID	Rule
BR-001	Only authenticated travelers may access TravelMate.
BR-002	Only Trip Owners may manage members.
BR-003	Invitations are sent via email only.
BR-004	Shared members must already have TravelMate accounts.
BR-005	Owners cannot remove themselves.
BR-006	Owners cannot change their own role.
BR-007	Traveler may update only their own profiles.
BR-008	Email address must remain unique.
BR-009	A traveler may participate in multiple trips.
BR-010	A trip shall have exactly one Owner.

7. Assumptions

- Users have internet connectivity.
- Email service is available.
- Members already have accounts.
- Travelers use supported web browsers and have stable internet connectivity.

8. Constraints

- Email is the only invitation channel.
- Guest access is not supported.
- Collaboration requires an existing trip.
- Existing member must register before they can be invited to a trip.

9. Traceability

BR ID	Business Requirement	Related FR IDs
BR-01	User Authentication	FR-001 – FR-007
BR-02	Trip Management	FR-008 – FR-013
BR-03	Collaboration	FR-014 – FR-022

10. Approval

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